

# Numerical Analysis Exercise 8

## 1. A natural cubic spline S on [0,2], is defined by

$$S(x) = \begin{cases} S_0(x) = 2x^3 - 3x + 4, & 0 \leq x \leq 1 \\ S_1(x) = a(x-1)^3 + b(x-1)^2 + c(x-1) + 3, & 1 \leq x \leq 2 \end{cases}$$

Find a,b and c.

**Solution:**  $S(1) = S_0(1) = 2 - 3 + 4 = 3 = S_1(1)$

Due to the natural cubic spline S, So we have the Solutions as follows:

$$\begin{cases} S_0''(0) = S_1''(2) = 0 \\ S_0''(1) = S_1''(1) = 12 \\ S_0'(1) = S_1'(1) = 1 \end{cases}$$

a = -2, b = 6, c = 3.

## 2. Curve fitting:

Find the power fits  $y = Ax^2$  and  $y = Bx^3$  for the following data and use  $E_2(f)$  to determine which curve fits better.

(1)

| $x_k$ | 2.0 | 2.3 | 2.6  | 2.9  | 3.2  |
|-------|-----|-----|------|------|------|
| $y_k$ | 5.1 | 7.5 | 10.6 | 14.4 | 19.0 |

```
1  clc;
2  clear all;
3  N = 5;
4  Sum1 = 0;
5  Sum2 = 0;
6
7  x = [2.0,2.3,2.6,2.9,3.2];
8  y = [5.1,7.5,10.6,14.4,19.0];
9  %y_k = [5.9,8.3,10.7,13.7,17.0];
10
11 x_2 = x.^2;
12 x_3 = x.^3;
13
14 syms A1 A2
15
16 %%Ax^2
17 for i = 1:N
18     Sum1 = Sum1+2*(A1*x_2(i)-y(i))*x_2(i);
19 end
```

```

20
21 sum1 = sum(Sum1,2);
22 A1 = solve(sum1,A1);
23 A1 = double(A1)
24 E2_1 = 0;
25 for i = 1:N
26     E2_1 = (A1*x_2(i)-y(i)).^2;
27 end
28 E2_1 = E2_1/N;
29 E2_1 = sqrt(E2_1)
30 %%Ax^3
31 for i = 1:N
32     Sum2 = Sum2+2*(A2*x_3(i)-y(i))*x_3(i);
33 end
34
35
36 sum2 = sum(Sum2,2);
37 A2 = solve(sum2,A2);
38 A2 = double(A2)
39 E2_2 = 0;
40 for i = 1:N
41     E2_2 = (A2*x_3(i)-y(i)).^2;
42 end
43 E2_2 = E2_2/N;
44 E2_2 = sqrt(E2_2)
45

```

```

1 A1 = 1.6866
2 E2_1 = 0.7734
3 A2 = 0.5902
4 E2_2 = 0.1512
5

```

It's easy find that the  $Ax^3$  better for (1).

(2)

| $x_k$ | 2.0 | 2.3 | 2.6  | 2.9  | 3.2  |
|-------|-----|-----|------|------|------|
| $y_k$ | 5.9 | 8.3 | 10.7 | 13.7 | 17.0 |

```

1 A1 = 1.6177
2 E2_1 = 0.1942
3 A2 = 0.5606
4 E2_2 = 0.6125

```

By using the code above, It's easy find that the  $Ax^2$  better for (2).

## 2.

Find the power fits  $y = \frac{A}{x}$  and  $y = \frac{A}{x^2}$  for the following data.

(1)

|       |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|
| $x_k$ | 0.5 | 0.8 | 1.1 | 1.8 | 4.0 |
| $y_k$ | 7.1 | 4.4 | 3.2 | 1.9 | 0.9 |

```
1  clc;
2  clear all;
3  Sum1 = 0;
4  Sum2 = 0;
5
6  x = [0.5,0.8,1.1,1.8,4.0];
7  x = 1./x;
8  %y = [5.1,7.5,10.6,14.4,19.0];
9  y = [7.1,4.4,3.2,1.9,0.9];
10 x_2 = x.^2;
11
12 syms A1 A2
13 N = size(x,2);
14 %%A/x
15 for i = 1:N
16     Sum1 = Sum1+2*(A1*x(i)-y(i))*x(i);
17 end
18
19 sum1 = sum(Sum1,2);
20 A1 = solve(sum1,A1);
21 A1 = double(A1)
22 E2_1 = 0;
23 for i = 1:N
24     E2_1 = (A1*x_2(i)-y(i)).^2;
25 end
26 E2_1 = E2_1/N;
27 E2_1 = sqrt(E2_1)
28
29 %%A/x^2
30 for i = 1:N
31     Sum2 = Sum2+2*(A2*x_2(i)-y(i))*x_2(i);
32 end
33
34 sum2 = sum(Sum2,2);
35 A2 = solve(sum2,A2);
36 A2 = double(A2)
37 E2_2 = 0;
38 for i = 1:N
39     E2_2 = (A2*x_2(i)-y(i)).^2;
40 end
41 E2_2 = E2_2/N;
42 E2_2 = sqrt(E2_2)
43
```

```
1 | A1 = 3.5339
2 | E2_1 = 0.3037
3 | A2 = 2.0060
4 | E2_2 = 0.3464
```

the power fits is  $f_1 = \frac{3.5339}{x}, f_2 = \frac{2.006}{x^2}$

(2)

|       |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|
| $x_k$ | 0.7 | 0.9 | 1.1 | 1.6 | 3.0 |
| $y_k$ | 8.1 | 4.9 | 3.3 | 1.6 | 0.5 |

```
1 | A1 = 4.6013
2 | E2_1 = 0.0050
3 | A2 = 3.9755
4 | E2_2 = 0.0261
```

the power fits is  $f_1 = \frac{4.6013}{x}, f_2 = \frac{3.9755}{x^2}$ .